

are not perfectly positively correlated with each other. Assumption 5 is used to guarantee that degeneracy in the asymptotic distribution of  $\gamma' \tilde{Y}_n$  arises only from known degeneracies in  $A$ . We note, however, that Assumption 5 does not rule out settings where the solutions to the bounds of the identified set given in equation (5) and (6) are non-unique or degenerate (*i.e.* where the extreme points for  $\theta$  occur at “flat faces” of the identified set). If  $A$  is full-rank, for example, then Assumption 5(A) holds trivially with  $T = I$ , and thus the mean of the moments  $\tilde{\mu}(\theta)$  is completely unrestricted.<sup>20</sup>

The assumptions stated above are sufficient for the conditions in Proposition 2 in ARP, which establishes uniform size control for the conditional and hybrid tests.

**Proposition 3.1.** *Suppose Assumptions 2–5 hold. Then the conditional and LF-hybrid tests uniformly control size. That is, for any  $\alpha < 0.5$ ,*

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} \mathbb{E}_P \left[ \psi_\alpha^C \left( \hat{\beta}_n, A, d, \theta_P, \frac{1}{n} \hat{\Sigma}_n \right) \right] \leq \alpha.$$

$$\limsup_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} \mathbb{E}_P \left[ \psi_{\kappa, \alpha}^{C-LF} \left( \hat{\beta}_n, A, d, \theta_P, \frac{1}{n} \hat{\Sigma}_n \right) \right] \leq \alpha.$$

### 3.4. Uniform asymptotic consistency

We next provide conditions under which the conditional and hybrid tests are uniformly asymptotically consistent, in the sense that power against fixed alternatives outside the identified set converges uniformly to 1. To establish uniform consistency of the conditional and hybrid tests, we strengthen Assumptions 2 and 3 as follows.

**Assumption 6.** *Let  $W_n = ((\hat{\beta}_n - \beta_P)', (\text{vec}(\hat{\Sigma}_n) - \text{vec}(\Sigma_P))')'$ , where  $\text{vec}(\Sigma)$  is the vector of the elements of the matrix  $\Sigma$ . We assume*

$$\lim_{n \rightarrow \infty} \sup_{P \in \mathcal{P}} \sup_{f \in BL_1} \left| \mathbb{E}_P [f(\sqrt{n} W_n)] - \mathbb{E} [f(\zeta_P^+)] \right| = 0,$$

where  $\zeta_P^+ \sim \mathcal{N}(0, V_P)$ ,  $V_P = \begin{pmatrix} \Sigma_P & V_{P, \beta \Sigma} \\ V_{P, \Sigma \beta} & V_{P, \Sigma} \end{pmatrix}$ .

**Assumption 7.** *For all  $P \in \mathcal{P}$ ,  $\Sigma_P \in \mathbf{S}$  and the matrix  $V_P$  defined in Assumption 6 lies in a compact set  $\mathbf{V}$ . Additionally,  $(\Sigma_P - V_{P, \beta \Sigma} V_{P, \Sigma}^\dagger V_{P, \Sigma \beta})$  has eigenvalues bounded below by  $\tilde{\lambda} > 0$ , where  $^\dagger$  denotes the Moore–Penrose inverse.*

Assumption 6 strengthens Assumption 2 to require that  $\hat{\beta}_n$  and  $\hat{\Sigma}_n$  have a joint normal asymptotic distribution. Although somewhat more restrictive, event-study estimates are often estimated via OLS, and standard covariance estimators for OLS, including cluster-robust variance estimators, produce asymptotically normal estimates as the number of clusters grows large (Hansen, 2007; Stock and Watson, 2008; Hansen and Lee, 2019). We do not impose that the asymptotic distributions of  $\hat{\beta}_n$  and  $\hat{\Sigma}_n$  are independent, as would occur in linear models if the linear model is correctly specified. Assumption 7 strengthens Assumption 3 to require that the asymptotic distribution of  $\hat{\beta}_n$  is not perfectly asymptotically colinear with  $\hat{\Sigma}_n$ .

20. Some values of  $A$  satisfying Assumption 5 may imply that certain pairs of moments cannot simultaneously be binding. For example, the restriction that  $|\delta_1| \leq 1$  can be represented as  $\delta_1 \leq 1$  and  $-\delta_1 \leq 1$ , which satisfies Assumption 5(A), but clearly both moments cannot simultaneously bind. Nevertheless Assumption 5(A) is compatible with “flat faces” even when  $A$  is not full rank. For example, if  $\Delta$  corresponds with the restrictions  $|\delta_1| \leq 1$  and  $|\delta_2| \leq 1$ , then the extreme points for  $\theta = \tau_2$  occur at flat faces of the identified set for  $(\tau_1, \tau_2)$ .

Under the imposed assumptions, we obtain uniform consistency of the conditional and hybrid tests.

**Proposition 3.2.** *Suppose Assumptions 4–7 hold. Then for any  $x > 0$  and  $\alpha < 0.5$ ,*

$$\lim_{n \rightarrow \infty} \inf_{P \in \mathcal{P}} \mathbb{E}_P \left[ \psi_\alpha^C \left( \hat{\beta}_n, A, d, \theta_P^{ub} + x, \frac{1}{n} \hat{\Sigma}_n \right) \right] = 1$$

$$\lim_{n \rightarrow \infty} \inf_{P \in \mathcal{P}} \mathbb{E}_P \left[ \psi_{\kappa, \alpha}^{C-LF} \left( \hat{\beta}_n, A, d, \theta_P^{ub} + x, \frac{1}{n} \hat{\Sigma}_n \right) \right] = 1,$$

where  $\theta_P^{ub} = \sup \mathcal{S}(\beta_P, \Delta)$  is the upper bound of the identified set. The analogous result holds replacing  $\theta_P^{ub} + x$  with  $\theta_P^{lb} - x$  for  $\theta_P^{lb} = \inf \mathcal{S}(\beta_P, \Delta)$ .

### 3.5. Optimal local asymptotic power

We next provide conditions under which the conditional test has optimal local asymptotic power. We first state the conditions and our formal results, and then provide several examples highlighting when the assumptions will and will not hold.

**3.5.1. Main results.** We begin by defining LICQ. Recall that the upper bound of the identified set is given by

$$\theta^{ub}(\beta, \Delta) = l' \beta_{post} - \left( \min_{\delta} l' \delta_{post}, \text{ s.t. } A\delta \leq d, \delta_{pre} = \beta_{pre} \right).$$

Since  $\delta_{post} = \beta_{post} - \tau_{post}$ , we can re-write the upper bound as a maximization over  $\tau_{post}$ ,

$$\theta^{ub}(\beta, \Delta) = \max_{\tau_{post}} l' \tau_{post}, \text{ s.t. } -A_{(\cdot, post)} \tau_{post} \leq d - A\beta, \quad (16)$$

where  $A_{(\cdot, post)}$  contains the columns of  $A$  corresponding with  $\delta_{post}$ . Let  $\tau_{post}^*$  denote a solution to the optimization for  $\theta^{ub}(\beta, \Delta)$  in (16), and let  $B^*$  denote the indices of the binding constraints, so that  $-A_{(B^*, post)} \tau_{post}^* = d_{B^*} - A_{(B^*, \cdot)} \beta$  and  $-A_{(-B^*, post)} \tau_{post}^* < d_{-B^*} - A_{(-B^*, \cdot)} \beta$ .

**Definition 2 (LICQ).** We say that LICQ holds in direction  $l$  if there exists a solution  $\tau_{post}^*$  to (16) such that the gradient of the binding constraints with respect to  $\tau_{post}$ ,  $-A_{(B^*, post)}$ , has full row rank.<sup>21</sup> We define LICQ in the direction  $-l$  analogously for the optimization that replaces max with min in (16).

For  $\epsilon > 0$ , we define  $\mathcal{P}_\epsilon$  to be the set of distributions  $P \in \mathcal{P}$  such that LICQ holds in the direction  $l$  and the non-binding constraints are slack by at least  $\epsilon$ , i.e.  $-A_{(-B^*, post)} \tau_{post}^* < d - A\beta_P - \epsilon$ .

Our next result states that for  $P \in \mathcal{P}_\epsilon$ , the local power of the conditional test converges to the power envelope for tests that control size in the finite sample normal model. To state this result formally, we define  $\mathcal{I}_\alpha(\Delta, \Sigma_n)$  to be the collection of confidence sets that control size in the finite sample normal model, i.e. confidence sets satisfying (10).

21. The definition of LICQ in Kaido *et al.* (2021) would require that this condition holds for all solutions  $\tau_{post}^*$ . For our results, however, it is sufficient for the condition to hold for some solution  $\tau_{post}^*$ .

**Proposition 3.3.** Suppose Assumptions 2–4 hold. Let  $\theta_P^{ub} = \sup \mathcal{S}(\beta_P, \Delta)$ . Then for any  $\epsilon > 0$ ,  $x > 0$ , and  $\alpha < 0.5$ ,

$$\lim_{n \rightarrow \infty} \sup_{P \in \mathcal{P}_\epsilon} \left| \mathbb{E}_P \left[ \psi_\alpha^C \left( \hat{\beta}_n, A, d, \theta_P^{ub} + \frac{1}{\sqrt{n}}x, \frac{1}{n}\hat{\Sigma}_n \right) \right] - \rho_\alpha^*(P, x) \right| = 0,$$

where

$$\rho_\alpha^*(P, x) = \lim_{n \rightarrow \infty} \sup_{C_{\alpha,n} \in \mathcal{I}_\alpha(\Delta, \frac{1}{n}\Sigma_P)} \mathbb{P}_{\hat{\beta}_n \sim N(\beta_P, \frac{1}{n}\Sigma_P)} \left( \left( \theta_P^{ub} + \frac{1}{\sqrt{n}}x \right) \notin C_{\alpha,n} \right)$$

is the optimal local asymptotic power of a size- $\alpha$  test in the finite sample normal model. An analogous result holds for the lower bound under the class of distributions where LICQ holds in direction  $-l$ .

Since the LF-hybrid test rejects whenever the conditional test with size  $(\alpha - \kappa)/(1 - \kappa)$  rejects, it is immediate that the local asymptotic power of the LF-hybrid test is at least as good as the power of the optimal size- $((\alpha - \kappa)/(1 - \kappa))$  test.

**Corollary 3.1.** Under the conditions of Proposition 3.3,

$$\liminf_{n \rightarrow \infty} \inf_{P \in \mathcal{P}_\epsilon} \left( \mathbb{E}_P \left[ \psi_{\kappa,\alpha}^{C-LF} \left( \hat{\beta}_n, A, d, \theta_P^{ub} + \frac{1}{\sqrt{n}}x, \frac{1}{n}\hat{\Sigma}_n \right) \right] - \rho_{(\alpha-\kappa)/(1-\kappa)}^*(P, x) \right) \geq 0.$$

We emphasize that Proposition 3.3 and Corollary 3.1 are new, and exploit structure in our context not contained in the more general setting considered in ARP.

**3.5.2. Discussion and examples.** As discussed in Kaido *et al.* (2021), LICQ and related constraint qualifications have been used frequently in the partial identification literature. Intuitively, LICQ ensures that the bounds of the identified set are differentiable with respect to the means of the moments  $(\tilde{\mu}(\tilde{\theta}))$ , and thus avoids challenges related to estimation and inference for non-differentiable parameters (Hirano and Porter, 2012). Uniform LICQ conditions have been invoked recently by Gafarov (2019) and Cho and Russell (2019), and a related Slater constraint qualification is used in Kaido and Santos (2014). One important distinction between our results and previous results using LICQ is that we do not require LICQ for our size control results (Proposition 3.1). Thus, our tests control size even when LICQ fails (and so the bounds may be non-differentiable), but Proposition 3.3 shows that this does not come at the cost of power asymptotically when indeed LICQ holds.<sup>22</sup>

Figure 1 provides geometric intuition for when LICQ will and will not hold in the case where  $\bar{T} = 2$  and the target parameter is the average of the post-treatment effects,  $\theta = \frac{1}{2}(\tau_1 + \tau_2)$ . In panel (a), there is a unique  $\tau_{post}^*$  (coloured in red) at which two linearly independent moments bind, so LICQ is satisfied. LICQ is likewise satisfied in panel (b), where the optimal  $\tau_{post}^*$  is not unique (a so-called “flat-face” problem). This is because at the indicated values  $\tau_{post}^*$  (coloured in red), there is either one or two linearly independent binding moments. A failure of LICQ is shown in panel (c). In this example, there are three binding moments at  $\tau_{post}^*$  (coloured in red), so the binding constraints cannot be linearly independent in  $\mathbb{R}^2$ . Such a situation may arise when there are both smoothness restrictions and sign or shape restrictions that are simultaneously binding at the boundary of the identified set.

22. We view this result as loosely parallel to results in the weak identification literature showing that certain procedures control size under weak identification but are efficient under strong identification (e.g. Moreira, 2003).

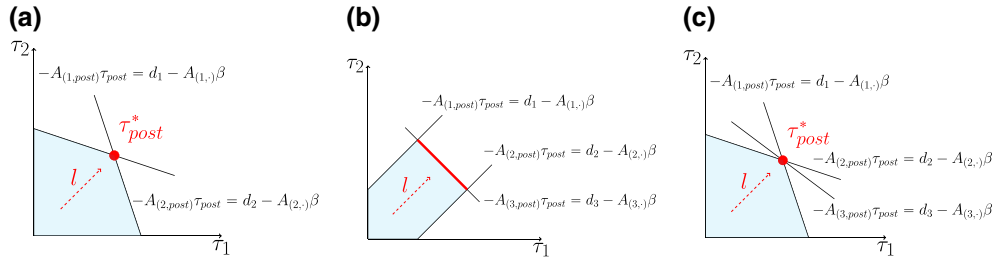


FIGURE 1

Diagram illustrating when LICQ (Assumption 2) will and will not hold in the case where  $\bar{T} = 2$ .

*Note:* In each panel, we assume that the rows associated with binding moments are ordered first in the matrix  $A$  for ease of notation. The blue shading denotes the identified set for  $(\tau_1, \tau_2)$  and the dashed red arrow points in the direction  $l = (\frac{1}{2}, \frac{1}{2})'$ . In panel (a), LICQ is satisfied since there is a unique  $\tau_{post}^*$  (coloured in red) at which two linearly independent moments are binding. In panel (b), even though  $\tau_{post}^*$  is not unique, LICQ is satisfied as there is either one or two linearly independent binding moments at the values of  $\tau_{post}^*$  coloured in red. In panel (c), there are three binding moments at  $\tau_{post}^*$  (coloured in red), and so LICQ is violated.

In the three period DiD model (where there are no nuisance parameters, since  $\bar{T} = 1$ ), LICQ is satisfied when the bounds of the identified set are each determined by one moment. This holds everywhere for  $\Delta^{\text{SD}}(M)$  when  $M > 0$ . It holds almost everywhere for  $\Delta^{\text{SDPB}}(M)$  when  $M > 0$ , although it fails when both the sign restrictions and smoothness restrictions are simultaneously binding. (For LICQ to hold with non-binding moments slack by at least  $\epsilon$ , i.e.  $P \in \mathcal{P}_\epsilon$ ,  $\delta_P$  must not be local to a point at which LICQ fails.) When  $M = 0$ , both the upper and lower bounds for  $\Delta^{\text{SD}}(M)$  and  $\Delta^{\text{SDPB}}(M)$  are binding, so LICQ fails.

More generally, the result in Proposition 3.3 is under the asymptotic regime where the sampling variation grows small relative to the length of the identified set, and thus the binding and non-binding moments are “far” apart relative to sampling variation. Importantly, it can be shown that the LICQ condition rules out settings where  $\theta$  is point identified. Thus, the asymptotics considered in Proposition 3.3 may not provide a good approximation to the finite-sample performance of the conditional test in settings where  $\theta$  is point-identified, or when the length of the identified set is “small” relative to sampling variation.

We are not aware of results analogous to Proposition 3.3 for any test that controls size in the finite-sample normal model. Kaido and Santos (2014) provide an efficiency result under a related Slater constraint qualification condition, but their test does not control size when the constraint qualification fails. It is worth highlighting that if LICQ holds for a particular set of moments, then it also holds if one adds moments that are slack at the optimal  $\tau_{post}^*$ . Proposition 3.3 thus requires that the asymptotic power of the test is not affected by the inclusion of slack moments. The only other non-trivial tests that we are aware of that control size in the finite-sample normal model and have this form of insensitivity to slack moments are the tests proposed by Cox and Shi (2022). An interesting open question is whether the tests proposed by Cox and Shi (2022) also converge to the power envelope under LICQ.<sup>23</sup>

**3.5.3. Extensions.** Proposition 3.3 is stated for the case when  $\Delta$  is a single polyhedron. An immediate corollary, however, is that when  $\Delta = \bigcup_{k=1}^K \Delta_k$ , the conditional test based on the union of confidence sets has optimal local asymptotic power when the  $\Delta_k$  that determines the identified set bounds is unique and satisfies the conditions of Proposition 3.3. This implies,

23. Extending the results to the Cox and Shi tests is non-trivial given that they use a different test statistic and construct critical values in a different way.